

SukiMart Finals — Confidence Playbook

Team SiXeven · UCLM · LAMBO 2026 Finals · LAMBO Main Stage, Waterfront
Cebu · June 27, 2026

What this is. The one document that holds *everything* you need to walk on stage confident and survive the Q&A — the case, the numbers (every one reconciled to our model), the business concepts in plain language, the full AI/system deep-dive, and a war-room of anticipated questions with crisp answers. Read it once cover-to-cover. Then drill **§9 (Q&A War Room)** and **§10 (Cheat Sheet)** until the numbers are reflexes.

The promise of this case (quote it to yourself): *"There is no prize for the most ambitious option. There is a prize for being right, and for being honest about the risk."* That sentence is our entire strategy.

Contents

Team SiXeven · UCLM · LAMBO 2026 Finals · LAMBO Main Stage, Waterfront Cebu · June 27, 2026

§0 · How to use this playbook

§1 · The case in 90 seconds

§2 · Our thesis (memorize this paragraph)

§3 · The facts (your reference shelf — all from the case exhibits)

Tina & the deal

Exhibit A — Today's P&L (the busiest store on the street)

Exhibit C — Cost to build SukiMart (one-time): ₱1,350,000

Exhibit D — SukiMart monthly fixed cost: ₱147,000

Exhibit E — Location ("real but not a flood")

Exhibit F — Convenience benchmarks & new category margins

Exhibit G — Operational detail (where the analysis points reward you)

Exhibit B — Today's sales mix (the buried lever)

§4 · Business crashcourse (the 8 ideas that decide the case)

§5 · The numbers, locked & reconciled

Today (baseline)

Break-even (SukiMart)

The three paths (consistent economics: economic profit = cash – ₱16,000 fair wage)

Loan math

AI savings — and why we cite the smaller number (rehearse this)

§6 · The recommendation & how to defend it

§7 · THE AI DEEP-DIVE — Smart-Prep (P4 owns this; everyone should know it)

7.1 What it is, in one line

7.2 Why spoilage (why AI earns its place here and not everywhere)

7.3 The method — two real models (this is why it's NOT "just an average")

7.4 The numbers it produces (backtested — conservative settings)

7.5 We tested ML — and said so honestly

7.6 The honest trade-off — the profit-vs-availability DIAL

7.7 Data quality — censoring (the detail that makes it real)

7.8 The data it needs & the tech stack (specifics for the "is it buildable?" question)

7.9 The horizon — one engine, compounding (maps to the case's Part 6!)

7.10 What we deliberately do NOT do (judgment, not omission)

7.11 Honesty / AI disclosure

§8 · The 12-month plan & risks

§9 · The Q&A War Room

Business & numbers

Strategy / the verdict

AI / the system (own these — P4 lead, all support)

Curveballs

§10 · One-page cheat sheet (the night-before page)

§11 · Appendix — reproduce every number yourself

§0 · How to use this playbook

Read order: §1 → §2 → §3 (context) in one sitting. Then your own section depending on who you present (§0 role map). Then §9 Q&A as a team, out loud, quizzing each other. Night before: §10 cheat sheet only.

Who owns what (4 presenters, 7:00 total):

You are	You own	Sections to master
P1 — Verdict	Opening + the recommendation + three-path comparison + close	§2, §6, §10
P2 — The Money	The ₱25,800 illusion, markup-vs-margin, the cost trap, the money slide	§4 (#1,#2,#4), §5
P3 — The Math & Plan	Break-even, location reality, the 12-month roadmap, risks	§4 (#3,#5,#6), §5, §8
P4 — Operations & AI	The store-builds-itself, Smart-Prep / the AI	§7 (whole), §9 AI block

The golden rule of the Q&A: answer in **one sentence with a number**, then stop. If you don't know, say *"I don't have that figure in front of me, but the way we'd compute it is..."* — judges reward method over bluffing.

§1 · The case in 90 seconds

Cristina "Tina" Mahinay, 44, has run **Tindahan ni Tina** in Barangay Guadalupe for 13 years — the busiest store on her street. About ₱10,000 passes through her hands on a good day. Her rule, for 13 years: *"buy ₱10,000 of goods, sell at 20% more — that 20% is my profit."* It fed her family, sent two kids to school, built **₱600,000 in savings**.

Now a branded 24-hour convenience store opened 1.4km away. Tina wants her own: **SukiMart** — air-conditioned, services, ready-to-eat, *"so my suki finally get the store they deserve."* A cooperative will lend her **₱750,000**; with her ₱600,000 she can build it. She was **ready to sign this week**.

Then her daughter **Jamie** (finishing a business degree) asked the question that stopped her:

"Ma. The twenty percent. Once we are paying rent, and aircon, and three staff, and the loan every month — is the twenty percent still enough? Or are we just busier and poorer?"

Tina has no answer. She has never paid rent (the store is her house), never paid herself a salary, never had a bill due on the 1st whether the store was busy or empty. **That is where we come in.**

The four questions we must answer (Part 5 of the brief): 1. **The verdict** — full jump, staged transition, or disciplined sari-sari? Decide, prove it. 2. **The break-even reality** — what sales & mix does SukiMart need to survive, then to profit? 3. **The role of AI** — where does AI earn its place, tied to a real problem, quantified in pesos or risk? *"Buzzwords score nothing."* 4. **The plan** — a 12-month roadmap, a financing/payback view, the three biggest risks + how she manages each.

Our answer in one line: *Don't bet the farm — earn the store.* Stage the transition; let the data earn the loan; use AI to kill the one cost that sinks convenience stores — spoilage.

§2 · Our thesis (memorize this paragraph)

"Tina's real profit is ₱9,800 a month, not the ₱25,800 she believes. SukiMart must earn ₱147,000 a month before she's paid — a 3.4× jump at a corner that's 'real but not a flood,' with six rival stores within 300 meters. So we don't bet her ₱600,000 savings and a ₱750,000 loan on day one. We add the high-margin pieces that fund themselves, prove the demand with data, and let AI kill the spoilage that would otherwise sink the highest-margin shelf — then take the loan when the numbers, not hope, say go."

Everything else in this playbook defends one of those sentences.

§3 · The facts (your reference shelf — all from the case exhibits)

Tina & the deal

- **₱600,000** personal savings (kept in the bank, never touched). **Loan offer:** ₱750,000 from a cooperative @ **18%/yr, 48 months** ≈ **₱22,000/mo** amortization.
- Today: **160 SKUs**, no staff, no digital records (a notebook). Convenience format would carry **~900 SKUs, 3 staff, perishables, a loan**.

Exhibit A — Today's P&L (the busiest store on the street)

Line	₱/mo	Note
Gross sales	180,000	~₱6,000/day, 96 transactions/day
Cost of goods sold	148,500	82.5% of sales
Gross profit	31,500	blended margin 17.5%
Electricity (store share)	2,800	hidden in the household bill
Spoilage + unrecovered utang	1,700	spoiled bread/ice, expired sachets, unpaid credit
Bags & supplies	800	
Load/e-wallet float & fees	400	
Cash left for Tina	25,800	← what Tina <i>thinks</i> she earns
(-) Fair owner wage	(16,000)	she works ~16 hrs/day for ₱0
= TRUE economic profit	9,800	← the real baseline SukiMart must beat

Exhibit C — Cost to build SukiMart (one-time): **₱1,350,000**

Renovation 380k · refrigeration/chillers 195k · shelving/gondolas 110k · POS+CCTV+network 95k · aircon 48k · signage 55k · initial inventory 280k · permits/BIR/deposits 92k · working-capital buffer 95k. **Financing:** ₱600k savings + ₱750k loan.

Exhibit D — SukiMart monthly fixed cost: **₱147,000**

Rent 28k · electricity 22k · staff 39k · **owner's salary 18k** · loan 22k · POS software 4k · permits 2.5k · maintenance 6k · marketing 3.5k · insurance 2k. *Note built into the data:* of the ₱22k loan payment, only the interest (~₱10k) is a true P&L expense; the principal (~₱12k) repays the balance sheet — **but all ₱22k still leaves the cash drawer**. Pick one view, be consistent.

Exhibit E — Location ("real but not a flood")

Barangay-road corner near a jeepney stop + elementary school; 220 sqm with the adjacent lot. **1,800 passersby/day · 2,400 households within 500m**. BUT **6 sari-sari within 300m**, a mid-size grocery 600m away, the branded 24h convenience store 1.4km away. Peaks: **6–8am (22%), 11–1 (16%), 4–7pm (28%)**.

Exhibit F — Convenience benchmarks & new category margins

Independent neighborhood convenience stores: **150–450 transactions/day, basket ₱90–140.** | Category | Margin | Note |
|---|---|---| | **Ready-to-eat** (rice meals, siomai, coffee) | **45–55%** | highest margin — **spoils fast if demand misjudged** | |
Chilled beverages | 18–22% | reliable traffic | | Snacks & confectionery | 25–30% | | Grocery & household | 12–16% | bulk
of the shelf, thin | | Tobacco | 8–10% | traffic driver, poor margin | | **Services** (bills, e-wallet, padala, load) | **fee-based ₱5–25/**
txn | near-pure margin; reason to walk in |

Warning in the data: unmanaged spoilage runs **8–15% of perishable sales**. Get forecasting wrong and the highest-margin category becomes the biggest loss.

Exhibit G — Operational detail (where the analysis points reward you)

- Hourly txn pattern is **uneven** (above) → staffing/forecasting signal.
- SKUs: **160 today** → **~900** convenience.
- **Utang ledger ₱12,000, of which 6% (~₱720) is never recovered.**
- Spoilage today concentrates in **bread, ice, slow sachets**.
- **No digital records.** *"A convenience store with 900 items, 3 staff, perishables, and a loan cannot be run from a notebook."*

Exhibit B — Today's sales mix (the buried lever)

Her **highest-volume** categories carry her **thinnest** margins: cigarettes 14% of sales @ **9%**, load 8% @ **8%**, soft drinks 16% @ 16%, rice 9% @ 10%. Her **fattest** margins sit in small lines: ice/misc 4% @ **30%**, sachet tingi 15% @ **28%**, bread 6% @ 25%, snacks 12% @ 22%. **Shifting the mix toward ready-to-eat + services is how the margin (and break-even) improves.**

§4 · Business crashcourse (8 ideas that decide the case)

For each: the idea, Tina's number, and the **on-stage line** to say it cleanly.

1 · Markup vs. Margin. Markup measures profit against **cost**; margin against **selling price** — margin is always smaller ($\text{margin} = \text{markup} \div (1 + \text{markup})$). Tina's "20% markup" is a **17.5% blended margin** (COGS is 82.5% of sales). *"Tina's '20%' is a markup on cost. On the selling price — how you run a P&L — it's a 17.5% margin. Every peso in this case is calculated on 17.5%, not 20%."*

2 · Fixed vs. Variable cost. Variable moves with each sale (COGS); fixed arrives regardless (rent, staff, loan). The case's whole trick: a sari-sari **hides** its three biggest fixed costs (rent ₱0 = her home, labor ₱0 = herself, electricity buried in the home bill). **Formalizing turns them into ₱147,000/mo of hard bills.** *"Sari-sari hides rent, labor, and electricity. SukiMart turns all three into bills due on the 1st — ₱147,000 a month that must be earned before Tina takes anything home."*

3 · Contribution margin & break-even. Break-even sales = $\text{fixed cost} \div \text{margin}$. SukiMart: $\text{₱147,000} \div 24\% = \text{₱612,500/mo} \approx \text{₱20,400/day} \approx 170 \text{ txns}$ at a ₱120 basket — vs **96 txns today**. A **3.4× jump**. The margin she hits decides whether that line is reachable:

Blended margin	Break-even/mo	Per day
17.5% (today's mix)	₱840,000	₱28,000
22%	₱668,000	₱22,300
24%	₱612,500	₱20,400
28% (pure P&L)	₱525,000	₱17,500

"SukiMart's fixed nut is ₱147,000. At a realistic 24% convenience margin she must sell ₱612,000 a month — 3.4× today — just to break even."

4 · Owner's wage vs. economic profit (the ₱25,800 illusion). A business hasn't earned anything until it pays for **all** inputs — including the owner's labor. Pay Tina a fair **₱16,000** wage and today's true profit is **₱9,800/mo**, not ₱25,800. (SukiMart does pay her an ₱18k salary — so even break-even SukiMart pays her more — but via ₱750k debt and ₱147k/mo of risk.) *"Tina thinks she earns ₱25,800. As a business paying a fair ₱16,000 wage, she earns ₱9,800. That's the number SukiMart must beat — and beat by enough to justify the risk."*

5 · Cash view vs. P&L view (the loan trap). Loan payment = interest (true expense on P&L) + principal (repays balance sheet, but still leaves the drawer). $\text{₱22k/mo} \approx \text{₱10k interest} + \text{₱12k principal}$. **Run survival on the CASH view** (all ₱22k → fixed cost ₱147k); a pure P&L would be ~₱135k. Pick one, state it. *"We run survival on cash — all ₱22,000 of the loan leaves her drawer, so break-even is ₱147,000. On a pure P&L only the ₱10,000 interest counts. A store survives on cash, so we lead with cash."*

6 · Sales mix. Blended margin is a **weighted average** of what you sell. Today her bestsellers are her worst margins. The convenience for sale of snacks ready-to-eat (45–55%) and services (near-pure) — **that mix shift is what makes ₱612k reachable.** *"Her margin isn't fixed — it's a weighted average. Today her bestsellers, cigarettes and load, are her worst margins. Ready-to-eat at 50% and services at near-pure margin are what move the break-even."*

7 · Payback & ROI. Payback = investment $\div$ annual gain. Full SukiMart: ₱1.35M capital; **base case** ~₱27,800/mo economic profit (a real return if base happens, ~10% payback); **pessimistic** ~₱46,000/mo and ~₱389k extra liquidity needed — **no payback**. The asymmetry is the risk. *"Base case, the full jump and our staged plan earn about the same — ₱27k a month. The difference is the downside: the full jump risks her ₱600k and can need ₱389k more in a bad year. So we earn the data first, then borrow."*

8 · Put it together (walk it like a judge). Real baseline ₱9,800 → must clear ₱147k fixed → ₱612k sales at 24% → ~170 txns/day → can a 'real-but-not-a-flood' corner with 6 rivals deliver that on day one? Base yes, downside ruinous → **stage it, prove it, then jump.** AI earns its place on the one quantified loss: **spoilage.**

§5 · The numbers, locked & reconciled

Every figure below reproduces from `SukiMart/analysis/sukimart_three_paths.py` and `SukiMart/smartprep/run.py`. If a judge challenges a number, this is your source of truth.

Today (baseline)

- Sales **₱180,000/mo** · GP **₱31,500 (17.5%)** · cash to Tina **₱25,800** · **economic profit ₱9,800** (after ₱16k fair wage).
- **96 transactions/day**, ₱6,000/day.

Break-even (SukiMart)

- Fixed **₱147,000/mo** ÷ 24% = **₱612,500/mo** ≈ **₱20,400/day** ≈ **170 txns @ ₱120**. 3.4× today.

The three paths (consistent economics: economic profit = cash – ₱16,000 fair wage)

Path	Mo-12 run-rate (econ profit/mo)	Year-1 cash to Tina (after capex)	Capital	Loan	Downside
3 · Disciplined sari-sari	₱27,200 (+63%)	₱353,232 ← highest Yr-1 cash	₱24–30k	No	≈ none
2 · Staged transition	₱27,200 (≈ 3×)	₱275,825	₱150k	No	≈ ₱150k capex; savings safe
1 · Full jump (base)	₱27,800	needs ₱1.35M upfront	₱1,350,000	₱750k	savings at risk; ~₱389k extra liquidity if sales low

Full-jump scenarios (why the loan is the gamble): | Scenario | Steady sales | Margin | Run-rate econ/mo | Yr-1 op cash | Downside liquidity | |---|---|---|---|---| | Pessimistic (150 txns/day) | ₱450,000 | 22% | **-₱46,000** | -₱483,750 | **-₱388,750** (needs ~₱389k) | | Base | ₱720,000 | 24% | +₱27,800 | +₱309,600 | +₱404,600 | | Optimistic | ₱1,092,000 | 25% | +₱128,000 | +₱1,386,750 | +₱1,481,750 |

The single most important insight: full-jump **base** run-rate (₱27,800) is **barely above** staged (₱27,200) — yet it costs **9× the capital and a ₱750k loan**. The full jump only *meaningfully* wins in its **optimistic** case (₱128k/mo). So the real question isn't "which earns more?" — it's "*is the optimistic case worth betting her ₱600k savings + a loan on, at a corner that's real but not a flood?*" **Answer: not until the data proves it.**

The honest trade-off (say it out loud): Staged gives up **~₱77,000 of Year-1 cash** versus Path 3 — because we reinvest into the ramp — *in exchange for* nearly 3× the run-rate, 12 months of demand data, and protection from an irreversible loan.

Loan math

₱750,000 @ 18%/yr, 48 months ≈ **₱22,000/mo** ≈ **₱10,000 interest** + **₱12,000 principal**.

AI savings — and why we cite the *smaller* number (rehearse this)

There are two ways to size the spoilage saving; **we deliberately pitch the conservative one**. - (a) **Top-down upper bound:** perishable revenue × the 7-point spoilage cut (16% → 9%). Staged ≈ ₱80k perishable → ~₱5,600/mo ≈ **~₱67k/yr**; full ≈ **~₱133k/yr**. *This is gross spoiled value, valued at revenue, ignoring lost-sale offset — an upper bound.* - (b) **Backtested, net (WHAT WE CITE):** from the actual model on sample data — **~₱1,983/mo** = **~₱24,000/yr staged**, **~₱47,000/yr full**. Lower



because it's **net of the sales given up by prepping leaner**, and spoilage is valued at **cost, not revenue**. *Our model's top-down ceiling is about ₱67,000 a year, but we pitch the backtested ₱24,000 — the net, proven figure. We'd rather under-promise and be right."*

§6 · The recommendation & how to defend it

Verdict: the Staged Transition (Path 2).

Why staged wins: 1. **Same run-rate as the full-jump base (₱27,200 ≈ ₱27,800) — with no loan and savings safe.** You capture the convenience upside without the irreversible bet. 2. **Each upgrade self-funds the next** — services (near-pure fee margin, draws traffic) → ready-to-eat (45–55%) → chillers — so she never bets money she hasn't earned. Her ₱180k store grows toward **₱320k/mo.** 3. **It generates the demand data** to answer "can this corner carry ₱612k/mo?" — so any future loan is **gated on proof, not hope.** The data is also what the AI runs on.

The rebuttals (drill these):

"Path 3 makes more cash in year one — why not just optimize the sari-sari?" "Path 3 wins year one by about ₱77,000, but it caps Tina at ₱16,000 a month and leaves the convenience opportunity on the table. Staged reaches a ₱27,000 run-rate and builds the data and muscle to go further. We trade a one-year cash bump for a durable, higher engine and a safe path to the full store."

"Aren't you just avoiding the real decision — the full build?" "No — we're sequencing it. Staged is the on-ramp. After 12 months we'll have proven demand and ₱450,000 of savings intact. If the data shows the corner carries ₱612,000 a month, Tina draws the loan THEN and builds from proof. Same destination, safer route."

"Isn't staging slower — a competitor could take the corner?" "The branded store is 1.4km away and the six rivals are sari-sari, not convenience. Tina's moat is 13 years of suki trust. Staging actually defends it — she adds what customers ask for, when the data confirms demand, instead of over-building and bleeding cash if the corner under-delivers."

§7 · THE AI DEEP-DIVE — Smart-Prep (P4 owns this; everyone should know it)

This is the section the judges said is worth real points and the section we lost on at PopCart. **It is built, backtested, and code-reviewed.** Reproduce any number with `./venv/bin/python SukiMart/smartprep/run.py`.

7.1 What it is, in one line

Smart-Prep is a feature inside Tina's POS that tells her each morning **exactly how many of each ready-to-eat item to cook** — sizing every item to its own economics so she stops wasting the highest-margin food.

7.2 Why spoilage (why AI earns its place *here* and not everywhere)

The case's Part 6 names six places AI could help. We picked the **one with a real, quantified loss and the data to learn from**: ready-to-eat earns the **best margin (45–55%) and bleeds the most — 8–15% spoilage** if demand is misjudged (Exhibit F). *"Buzzwords score nothing"* — so we costed one use, proved the payback, and mapped the rest as a data-earned roadmap (§7.9). **That discipline is the answer.**

7.3 The method — two real models (this is why it's NOT "just an average")

Step A — Forecast the demand (per item, per day). Holt-Winters exponential smoothing (a.k.a. triple exponential smoothing): it learns **level + trend + weekly seasonality (period = 7)** and adapts as patterns shift — plus a learned **payday multiplier** (15th/30th ± 1). It is *not* a fixed multiplier; it re-weights toward recent weeks automatically. - On short history (< 3–4 weeks) it falls back to a **seasonal baseline** (weighted average of recent same-weekdays $\times$ event factor). The model is **chosen per SKU by lowest hold-out error.**

Step B — Model the uncertainty. The decision needs the demand *distribution*, not just the mean: - High-volume items (coffee, pandesal): **Normal** (μ = forecast, σ = recent forecast-error). - Low-volume counts (rice meals, siomai): **Poisson** (λ = forecast); if variance > mean, **Negative Binomial**. (*Correct statistics for counts — and defensible.*)

Step C — The decision: the newsvendor model (the textbook operations-research method for perishables). Prepare the quantity where the cost of spoiling one more unit equals the cost of missing one more sale:

PREP $Q^* = F^{-1}(CR)$, where **$CR = C_u / (C_u + C_o)$** — C_u = margin lost on a stock-out (underage), C_o = cost lost on spoilage (overage). - Normal: **$Q^* = \mu + z \cdot \sigma$** , **$z = \Phi^{-1}(CR)$** . Poisson: **$Q^* = \text{poisson.ppf}(CR, \mu)$** .

The "smart part" — a different, costed decision per item: | Item | CR | Decision | ---|---|---| Coffee (high margin) | **0.60** | prep **above** the mean (a missed sale costs more than a spill) | | Pandesal (cheap to waste) | **0.40** | prep **below** the mean (spoilage more than a rare miss) | *Worked example (coffee, payday): $\mu \approx 32$, $CR = 18/(18+12) = 0.60$, $z \approx 0.25 \rightarrow Q^* \approx 33$ ("costed", not 34").*

The defensibility line: *"It's a payday-aware Holt-Winters forecast plus the newsvendor model — the standard method for perishables. A different, costed decision per item, and we backtested it. Not an average."*

7.4 The numbers it produces (backtested — conservative settings)

Metric	Result
Forecast accuracy (WAPE)	~19%
Spoilage of what's prepared	~16% (gut-feel) → ~9% (Smart-Prep) — ~44% less waste
Profit protected	~₱1,983/mo ≈ ₱24,000/yr staged → ~₱47,000/yr full
In-stock (service) level	71% → 53% at the profit-optimal setting — a <i>deliberate, tunable</i> trade-off
Incremental cost	₱0 — a feature of the POS already in the plan (Exhibit D)

7.5 We tested ML and said so honestly

Model comparison by average WAPE: **payday-Holt-Winters 18% · Holt-Winters 19% · LightGBM 20% · seasonal baseline 21%**. "We tried gradient boosting — LightGBM. At ~120 days of one-store data it scored 20%, worse than Holt-Winters' 18%. ML didn't earn its complexity yet, and we said so. When she has years of data, the framework promotes it automatically."

7.6 The honest trade-off — the profit-vs-availability DIAL

Smart-Prep is a **dial, not a fixed point**. At the profit-optimal setting it accepts more stock-outs of items (over-cooking pandesal wastes more than the occasional miss). Raise the service floor and you buy availability for profit: | Service floor | In-stock | Spoilage | Profit/yr (staged) | |---|---|---|---| | 0.00 (profit-optimal) | 53% | 9% | ₱24,000 | | 0.60 | 66% | 12% | ₱20,000 | | 0.70 | 74% | 15% | ₱14,000 | | 0.75 (≈ match today) | 79% | 17% | ₱10,000 | "When matching today's availability, it still cuts waste and adds profit — because it's day-aware, not flat. Tina turns one knob to trade availability for profit."

7.7 Data quality — censoring (the detail that makes it real)

On real POS data you only see what **sold**; on a sell-out, true demand was higher. In validation, sell-outs hit **33% of item-days**, and naive "sold" **under-counts demand by ~17%** on those days. Smart-Prep **de-censors** (using the sell-out time) and recovers demand to **within ~1%** — so the forecast doesn't spiral down. "If you train on 'sold,' a sell-out teaches the model to forecast less, and it spirals. We flag sell-outs and correct for them — that's the difference between a real system and a toy."

7.8 The data it needs & the tech stack (specifics for the "is it buildable?" question)

Five small tables — the one most teams forget is **prep_log** (it's the learning loop): - **items** (price, cost, shelf-life, perishable flag) · **sales** (POS export, one row/line) · **prep_log** (prepared / sold / spoiled / sold-out flag+time — 30 seconds at close) · **calendar** (dow, payday, holiday, school-day, weather) · **forecasts** (audit trail). - **Data needed: ~3–4 weeks** to learn weekly seasonality; cold-start uses category priors + Tina's gut with wide uncertainty (still beats guessing). - **Stack:** Python 3.11 · pandas/numpy · **statsmodels** (Holt-Winters) · **scipy.stats** (Poisson/NegBin/Normal quantiles) · lightgbm (later) · SQLite/CSV storage · openpyxl/Streamlit report · matplotlib. **Runs on a phone or any laptop in milliseconds — no cloud, no GPU.** - **Day-one reality:** the whole MVP runs in **Google Sheets** (the seasonal baseline + newsvendor are spreadsheet formulas — that's the **SukiMart_SmartPrep.xlsx** we built); production is the POS exporting CSV nightly to the engine, pushing the prep list to her phone each morning.

7.9 The horizon — one engine, compounding (maps to the case's Part 6!)

Smart-Prep's spoilage forecast is the beachhead, not the ceiling. The same forecast-then-optimize engine, fed by the POS data she captures from Month 1, extends to the *other* openings the case itself listed — each switching on as data arrives, **none needing a new system:** | Case Part-6 opening | Our extension | Same-engine logic | Status | |---|---|---|---| | "inventory & replenishment problem" (900 SKUs) | **Smart Reorder** | newsvendor reorder-point per SKU (`reorder.py` exists) | designed · code exists | | "loyalty & customer problem" (suki + utang) | **Utang scoring** | classify suki by repayment history → safe credit line (cuts the ₱720/mo never-recovered) | designed | | (raise the ₱62 basket toward ₱120) | **Basket lift** | market-basket cross-sell | designed | | "staffing & scheduling problem" (Exhibit G hours) | **Shift-to-traffic** | staff the 4–7pm peak, not the 9am–11am | designed | | "loss-prevention / shrinkage" | (later) | POS–inventory reconciliation flags | noted | | "go/no-go for opening" at a marginal location | = **our M9 gate** | the same demand model decides the loan | **in the plan** |

The unifying line: "Smart-Prep isn't 'the AI' — it's the first thing the AI does. Tina earns the right to grow the store with data, and that same data earns the right to grow the intelligence. One engine, fed not rebuilt."

7.10 What we deliberately do NOT do (judgment, not opinion)

No payday surge-pricing. No camera surveillance of customers. Both would erode the **suki trust** that is the business's real asset. Naming what we won't build is part of a mature answer. *"We could surge-price on payday. We won't — it would cost Tina the trust that took 13 years to build. We chose where AI belongs, and where it doesn't."*

7.11 Honesty / AI disclosure

Daily sales & sell-out timing in the prototype are **illustrative samples** — **the logic is the deliverable**. Only the **spoilage** use is built and backtested today; reorder/utang/baskets are designed extensions of the same engine, not yet running. AI tools (Claude/ChatGPT) assisted analysis, copy, and red-teaming; **Python** built the model; the prototype was **adversarially code-reviewed** and every number reconciles to `run.py`. **The founder always approves the final prep amount.**

§8 · The 12-month plan & risks

Roadmap (NO loan drawn this year): - **M1–2 · Baseline & books:** put the store on a cheap POS; start capturing sales/SKU/utang data (end the notebook). - **M3–5 · Services + ready-to-eat:** expand bills/e-wallet/padala; launch coffee + siomai; **switch Smart-Prep on** from the first day of perishables. - **M6–8 · Chillers + mix shift:** chilled beverages; push the mix toward fat-margin lines. - **M9 · ★ DECISION GATE:** does the data show a credible path to ₱612k/mo? **Go / refine / hold.** - **M10–12 · Prepare, don't borrow:** if the gate passes, assemble the loan package + full-build plan for **Year 2** (lease the adjacent lot, get 3 contractor quotes). **The ₱750k loan is not drawn inside this 12-month plan.** If the gate fails, bank a ~₱27k/mo store with savings intact.

The three biggest risks → mitigation → stop-rule: | Risk | Mitigation | Stop-rule | |---|---|---| | **Spoilage** (highest-margin line bleeds) | Smart-Prep forecast + start small | Cut the menu, not the savings | | **Demand never reaches break-even** | That's why we stage; the M9 gate | The gate stops the loan before it's taken | | **Owner overload** (1 person, 900 SKUs) | POS + one trained helper before scope grows | No scope growth until staffed |

Overarching stop-rule: if the economics turn negative, cut scope — don't borrow to chase it.

§9 · The Q&A War Room

Format: Q → **the one-line answer (lead with the number)**. Drill until reflexive. Categories: Business · Strategy · AI · Curveballs.

Business & numbers

1. **Is 20% markup or margin?** → Markup on cost; the margin on price is **17.5%**.
2. **What does Tina really earn?** → **₹9,800/mo** economic profit, not ₹25,800 (after a ₹16k fair wage).
3. **What's the break-even?** → ₹147k fixed ÷ 24% = **₹612,500/mo ≈ ₹20,400/day ≈ 170 txns**.
4. **How big a jump is that?** → **3.4×** today's ₹180k; **170 txns/day vs 96**.
5. **Cash or P&L view?** → **Cash** for survival (₹147k); P&L would be ~₹135k. We're consistent on cash.
6. **Where did ₹147k come from?** → Exhibit D: rent 28 + elec 22 + staff 39 + salary 18 + loan 22 + POS 4 + permits 2.5 + maint 6 + mktg 3.5 + insurance 2.
7. **What's the loan cost?** → ₹750k @18%/48mo ≈ **₹22k/mo** (~₹10k interest + ₹12k principal).
8. **Why 24% margin, not today's 17.5%?** → Mix shift to ready-to-eat (45–55%) + services (near-pure) lifts the blend; 24% is the conservative convenience benchmark (Exhibit F), not today's sari-sari mix.
9. **What's Tina's payback on the full jump?** → ~2 years *in the base case*; **none** in the pessimistic case — that asymmetry is why we stage.

Strategy / the verdict

1. **Why not the full jump?** → Base run-rate (₹27,800) ≈ staged (₹27,200), but it risks her ₹600k savings + a ₹750k loan on an unproven corner; it only truly wins in the optimistic case.
2. **Why not stay a disciplined sari-sari?** → It wins Year-1 cash by ~₹77k but **caps her at ₹16k/mo** and forfeits the convenience upside; staged reaches ₹27k *and* builds the data.
3. **Aren't you avoiding the real decision?** → No — sequencing it. Staged is the on-ramp; the M9 gate makes the loan a proof-based decision in Year 2.
4. **What if a competitor takes the corner?** → The branded store is 1.4km away; the 6 rivals are sari-sari. Her moat is 13 years of suki trust; staging defends it by adding what customers confirm they want.
5. **What's the one number to remember?** → **₹612,000** — the monthly sales SukiMart must clear to survive.

AI / the system (own these — P4 lead, all support)

1. **Is this really AI, or just an average?** → Payday-aware **Holt-Winters + the newsvendor model** — both standard, both backtested. A costed decision *per item*, not an average.
2. **What forecasting model, exactly?** → Holt-Winters (triple exponential smoothing, weekly seasonality period 7) for the mean; Poisson/Negative-Binomial or Normal for the distribution; **newsvendor** for the quantity.
3. **What data does it need?** → Daily sales + a **prep_log** (prepared/sold/spoiled/sold-out) + a calendar (payday/holiday/weather). ~3–4 weeks to learn the weekly pattern; works from day one on priors.
4. **Did you try machine learning?** → Yes — LightGBM scored **20% WAPE**, worse than Holt-Winters' **18%** at ~120 days of one-store data. We deploy the simpler model and said so.
5. **How much does it save, and at what cost?** → ~**₹24k/yr** staged, ~**₹47k** at full scale, **₹0 incremental** — it's a feature of the POS she's buying anyway.

6. **Why does your in-stock drop to 53%?** → That's the *profit-max* setting, deliberately on cheap items where over-cooking wastes more than a rare miss. It's a **dial** — match today's ~79% availability and it still cuts waste and adds profit.
7. **Why only ₱24k when your model shows more?** → Our top-down ceiling is ~₱67k/yr, but we cite the **backtested, net ₱24k** — net of sales given up and valued at cost. We under-promise.
8. **Is AI overkill for a sari-sari?** → The forecasting runs in a **spreadsheet** in milliseconds and Tina reads one number per item. It's affordable and explainable — that's the point.
9. **Why only spoilage — is that all AI can do here?** → Spoilage is where AI pays *today*. It's **one engine**: it extends to reorder across 900 SKUs, utang scoring, basket lift, and staffing-to-traffic — exactly the openings the case lists — switching on as the data arrives.
10. **Can Tina actually run it?** → The system recommends; **Tina decides**. She reads the morning list; the engine is built once by a dev and runs nightly.
11. **What did you choose NOT to build, and why?** → Surge-pricing and customer surveillance — they'd erode the suki trust that is her real asset.
12. **How does the AI handle a sell-out?** → It flags it and de-censors (true demand $\geq$ sold); otherwise the forecast would spiral down. Naive data under-counts by ~17%; we recover to ~1%.

Curveballs

1. **What if she can't collect the prep_log discipline?** → Day-one it's a 30-second tally; the POS automates it later. Without it she can't measure spoilage at all — so it's the first habit we build (M1–2).
2. **What if the cooperative loan terms change?** → Doesn't affect Year 1 — we don't draw the loan. The M9 gate re-prices the decision with whatever terms exist then.
3. **Isn't ₱9,800 unfair — she's happy with ₱25,800 cash?** → For her *household* cash flow, ₱25,800 is real. But to compare against SukiMart (which pays her a salary), we must value her labor — otherwise we'd compare a job-plus-business to a business and mislead her.
4. **What's your confidence in the 3.4× being reachable?** → Honestly: base case yes, downside no — which is *precisely* why we refuse to bet the loan on it and instead measure it first.
5. **If you had to cut one thing from the plan, what?** → The chillers (M6–8) before services/ready-to-eat — services draw traffic and ready-to-eat carries the margin + the AI story; chillers are the most deferrable.

If you genuinely don't know: *"I don't have that exact figure in front of me — here's how we'd calculate it: ..."* Then give the method. Never invent a number.

§10 · One-page cheat sheet (the night-before page)

The 7 numbers (memorize cold): - **₱9,800** — true economic profit today (not ₱25,800) - **₱147,000** — SukiMart's monthly fixed nut - **₱612,000** — monthly break-even (≈ 170 txns/day; $3.4\times$ today's 96) - **₱1.35M / ₱750k** — full-jump capital / the loan (₱22k/mo) - **₱27,200 vs ₱27,800** — staged vs full-jump base run-rate ($\approx$ tie) - **~₱389k** — extra liquidity the full jump needs in the downside - **16% $\rightarrow$ 9%** — spoilage with Smart-Prep; **~₱24k/yr** protected, **₱0 cost**

The 5 killer lines: 1. *"Don't bet the farm — earn the store."* 2. *"Tina thinks she earns ₱25,800. As a business, she earns ₱9,800."* 3. *"Same destination as the full jump — a fraction of the risk."* 4. *"Smart-Prep isn't the AI; it's the first thing the AI does."* 5. *"No prize for the most ambitious option — a prize for being right."*

The AI in one breath: Holt-Winters forecast + newsvendor optimizer $\rightarrow$ cook the profit-optimal amount per item $\rightarrow$ spoilage 16% $\rightarrow$ 9%, ~₱24k/yr, ₱0 cost, backtested (it beat LightGBM). One engine that compounds into reorder, utang, and baskets.

Pitch flow: Illusion (₱9,800) $\rightarrow$ Trap (markup $\neq$ profit, ₱147k) $\rightarrow$ Mountain (₱612k, $3.4\times$) $\rightarrow$ Corner (real but not a flood) $\rightarrow$ 3 Paths (same profit, different risk) $\rightarrow$ Verdict (earn the loan) $\rightarrow$ Store builds itself $\rightarrow$ **AI/Smart-Prep** $\rightarrow$ The money $\rightarrow$ 12-month plan + gate $\rightarrow$ Risks $\rightarrow$ Close.

Two silences that land: after "₱9,800" and after "her life savings, gone."

§11 · Appendix — reproduce every number yourself

```
# Financial model (three paths, break-even, AI reconciliation)
./venv/bin/python SukiMart/analysis/sukimart_three_paths.py

# The AI system (forecast + newsvendor + backtest + model comparison + censoring)
./venv/bin/python SukiMart/smartprep/run.py

# The tests (12 pass)
./venv/bin/pytest SukiMart/smartprep/tests/ -q
```

Source docs: business concepts → [analysis/BUSINESS_CRASHCOURSE.md](#) · the verdict → [analysis/THREE_PATH_VERDICT.md](#) · AI pitch handoff → [smartprep/SMARTPREP_AI_SPEC.md](#) · AI engineering → [smartprep/SMARTPREP_BUILD.md](#) · the deck → [deck/SUKIMART_FINALS_DECK.md](#).

Prepared for Team SiXeven — LAMBO 2026 Finals. Every figure reconciles to the model. Go earn it.